

Technique for quantification of embodied carbon footprint of construction projects using probabilistic emission factor estimators



Zhiquan Yeo^{*}, Ruisheng Ng, Bin Song

Singapore Institute of Manufacturing Technology, 71 Nanyang Drive, Singapore 638075

ARTICLE INFO

Article history:

Received 28 August 2014

Received in revised form

29 September 2015

Accepted 25 January 2016

Available online 2 February 2016

Keywords:

Carbon footprint

Life-cycle assessment

Embodied emissions

Green building

Computational tools

ABSTRACT

Buildings consume a significant proportion of the world's resources, mostly in the form of materials usage. In addition, with the ongoing efforts focusing on reducing the operational energy consumption of buildings, the embodied emissions of buildings are expected to increase in proportion in the future. With the increasing focus placed on the embodied emissions of buildings, organizations in the construction industry will face requirements to report information regarding the embodied emissions of buildings. However, in order to obtain the embodied emission of buildings, tremendous efforts have to be placed as the process is data intensive. In order to overcome this challenge, a streamlined technique is proposed to minimize the efforts required by practitioner to obtain the embodied carbon footprint. The proposed technique comprises a probabilistic model of emission factor estimators used to estimate the required embodied emissions. Based on the four projects presented as case study for the proposed technique, the practitioner would be required to manually match the appropriate emission factors to the activity data for between 12 and 21% of the data points. This is in contrast to the current traditional technique, which requires the manual matching of 100% of data points. The resultant deviation of the computed embodied carbon footprint from the proposed technique and the current technique was between -0.25% and $+3.65\%$.

© 2016 Elsevier Ltd. All rights reserved.

1. Introduction

1.1. Background

According to the United Nations Environment Programme, buildings consume about 40% of the world's resources, mostly in the form of materials usage (UNEP, 2007). A large myriad of materials is used which include concrete, cement based products, timber, stones, ceramic tiles, glass, metals like steel, aluminum and copper, paint and coating and various types of plastics such as PVC, acrylic and PET. Large amounts of resources and energy are required when producing and recycling these materials. Each material has to be extracted, processed and finally transported to its place of use. More than 30 billion tons of various materials are used in buildings each year (Rincón, 2011). Correspondingly, in 2010 buildings accounted for 32% of total global final energy use and 19% of energy-related greenhouse gas (GHG) emissions. This energy use

and related emissions is expected to double or potentially even triple by mid-century (Lucon et al., 2014).

Mitigating the release of GHGs into the atmosphere has been a growing issue in the area of sustainability. Therefore, to limit the use of resources and environmental degradation, the concept of 'green building' is proposed to reduce the environmental impact of buildings throughout their life cycle. At present, materials selection in building design are based almost solely on materials and products that had been labeled or certified green in accordance with criteria set out in the respective schemes prevalent in the countries of origin or exported to. Such schemes include the Singapore Green Building Product Certification Scheme (SGBC, 2014) developed by the Singapore Green Building Council, Singapore Green Labeling Scheme (SEC, 2014) by the Singapore Environment Council and China Energy Conservation Program (CECP, 1998). Malaysia has SIRIM Certified Eco-labeling scheme (SIRIM, 1997) and IBU Environmental Product Declaration (IBU, 2004). Other commonly encountered and well-known schemes include the Good Environmental Choice Australia (GECA, 2002), Blue Angel in Germany (Blue Angel, 1978) and Energy Star (Energy Star, 1992) and Design for the Environment in the United States (EPA, 1992).

^{*} Corresponding author.

E-mail address: zqyeo@SIMTech.a-star.edu.sg (Z. Yeo).

The usage of energy is closely linked to the release of greenhouse gases as approximately 80% of the global primary energy supply is derived from carbon-based fossil fuel (International Energy Agency, 2013). Energy and pollutant emissions such as carbon dioxide (CO₂) may be regarded as being 'embodied' within materials. That is to say, embodied energy of primary production can be viewed as quantity of energy required to extract, process, manufacture and transport the materials that constitute the building (Koskela, 1992; Lee et al., 2011; Bribián et al., 2011; Sahagun and Moncaster, 2012). Gonzalez and Navarro (González and Navarro, 2006) asserted that building materials possessing high embodied energy could possibly result in higher carbon dioxide emissions in comparison to materials possessing low embodied energy. Recently, under the EU Framework Program Horizon 2020, researchers carried out work to study the behavior of eco-building materials and the ecological significance of the low-carbon materials (Torgal, 2014). It was shown that in the assessment of the ecological performance of building materials, the structure or constitution of energy consumption and the methodology for evaluation are the main factors to be taken into consideration (Lee et al., 2011; Bribián et al., 2011). Various studies show that, for buildings, embodied carbon footprint is 37–43% of a 60 year life cycle carbon emissions in the UK (Eaton and Amaton, 1998), 11–50% of a 60 year life cycle in the USA and Canada (Engin and Frances, 2009), 45% of 50 years life cycle in Sweden (Thormark, 2002), and 60% of 50 year life cycle in Israel (Huberman and Pearlmutter, 2008). This shows that the traditional views to ignore embodied energy and the associated carbon emissions are not appropriate (Langston and Langston, 2008; Holtzhausen, 2007). This is particularly true due to the emerging trends towards "zero energy" buildings in which renewable energy coupled with low-energy consumption measures resulting in close to zero energy consumption derived from fossil fuels. As buildings become more energy efficient over time, the level of the energy consumption of the building is set to decrease in the foreseeable future. This will result in rising significance placed on the embodied energy and the associated carbon emissions of buildings (Plank, 2008; Nabel et al., 2008; Sartori and Hestnes, 2007; Optis and Wild, 2010; Ibn-Mohammed et al., 2013).

1.2. Life-cycle assessment for environmental evaluation of buildings

Life-cycle assessment (LCA) method have been used for environmental evaluation of product development processes in other industries for a long time (Cole and Rousseau, 1992; Jönsson, 1998; Suzuki and Oka, 1998), although application to the building construction sector has only been adopted for the last 10 years (Singh et al., 2011; Buyle et al., 2013). LCA is a tool for systematically analyzing environmental performance of products or processes over their entire life cycle, including raw material extraction, manufacturing, use, and end-of-life (EOL) disposal and recycling (Silvestre et al., 2014). Hence, LCA is often considered the most suitable approach to the evaluation of environmental impacts.

Efforts have been made to apply the LCA to the assessment and design of green buildings. Design of green buildings is achieved by reducing its operating energy through active and passive technologies. Abanda et al. (2013) presented a comprehensive set of 25 models for computing the embodied energy, GHG emissions and cost associated to construction activities. Multi-criteria evaluation models have also been established to evaluate material choices of buildings especially when conflicting objectives exist (Akadiri et al., 2013). Approaches in reducing buildings' operating energy could include changing the source of energy (e.g. from coal to natural gas), improving efficiency in equipment that utilize energy, and using green-materials that are also high performance. However, reduction in operating energy may sometimes be accompanied by an increase

in embodied emissions of the building due to the energy intensive materials used to achieve energy saving (Ramesh et al., 2010). Generally, as the operating energy decreases, the embodied emission as a proportion of the building's life cycle emissions increases. Thormark (2002) reported that embodied energy and its share in the life cycle energy for low energy building is higher than conventional ones. At present, the application of LCA to the green buildings can be divided into three aspects: (1) the quantitative evaluation of energy consumption and environmental emission, which can provide guidance for the ecological design and rehabilitation; (2) comparison between more than two kinds of materials by considering some performance requirements to give advice when selecting building materials; (3) analysis on the total energy consumption and emission of the building itself throughout the life cycle.

1.3. Challenges

1.3.1. Rising requirement for quantification of embodied emissions

With the increasing focus placed on the embodied emissions of buildings, organizations in the construction industry will face requirements to report information regarding the embodied emissions of buildings. For example, Singapore's latest version of the green building scheme Green Mark 4.0 (BCA, 2012a, 2012b) has started to include the carbon footprint as one of the assessment criteria, which requires the measurement of the embodied carbon footprint attributed to the materials used in construction for residential and non-residential buildings.

Organizations in the construction industry embarking on carbon footprint studies, however, face major hurdles in the process of measuring the embodied carbon footprint for building projects. Langston and Langston (2008) suggested that, while measuring operating energy is less complicated, determining embodied energy is more complex and time consuming. This is due to the time consuming process of the data collection and data processing efforts to collate the bill-of-materials (BOM) of the buildings and convert them into their corresponding embodied emissions.

1.3.2. Practical challenges in data collection

Various surveys conducted on LCA practitioners have similarly highlighted that data collection phase is the biggest challenge commonly faced when conducting life cycle studies. Cooper and Fava (2006) state that approximately two-thirds of LCA practitioners cited that the inventory data collection process is the most time-consuming and costly part of LCA. Subsequent surveys done by Teixeira and Pax (2011) also presented similar results on the challenges of conducting LCA. Previous studies have proposed carbon footprint streamlining methods applied in other context such as data grouping suggested by Rugrungruang et al. (2010) can be done on electronic products (Rugrungruang et al., 2010) and algorithmic approach by Ng et al. (2012) to streamline carbon footprint quantification for metal stamping production operations. Predictive emission factors in combination with structured data and uncertainty analysis were also developed to deal with tremendous data requirements for simultaneous multi-product carbon footprinting (Meinrenken et al., 2012). Yeo et al. (2013) have established a method combining activity data screening, emission estimation and cut-off criteria for fast approximation of carbon footprint results.

Issues arising from the process of data collection must first be understood in order to streamline the process of deriving the embodied carbon footprint for buildings. One of the largest challenges associated to data collection, is the availability of timely and complete data (Reap et al., 2008). The size and complexity of dataset often compounds the issue. Tremendous practical challenges exist due to the decentralized nature of the activity data

sources. These data sources are commonly distributed among several individual sub-contractors. As a result, the raw activity data required to build up the life cycle inventory is not available instantaneously. Hence, it is common for practitioners to have only acquired only a partial set of data after several iterations of data collection from the data sources. On top of that, there is a lack of standardization within the building sector with regard to data inventory as datasets are based on local and simple materials but not component or composites (Khasreen et al., 2009). In order to overcome this issue, it may be required to disaggregate an item's weight into its major constituent materials and specific composition. Additionally, uncertainties in the other life cycle stages of buildings such as the operating phase exist. Complexities arise due to the numerous operating scenarios that may occur and change over the life span of the building. In the absence of such detailed information prior to the completion and operation of the building, it is not possible to compute the carbon footprint results due to such data gaps, leading to delayed delivery of the results and findings of the carbon footprint study.

1.3.3. Data quality requirements

Product carbon footprint standards governs that all compliant carbon footprint studies are ensured through their guiding principles of consistency, transparency, completeness, relevance and accuracy (BSI, 2011; WRI, 2011; ISO, 2013). When the standards are applied to products consisting of a small number of components, effort required in collecting and processing data are often manageable. For instance, the quantification of the product carbon footprint of a paper cup can be completed in a relatively short amount of time as compared to a more complex product such as a building which consists of a large number of components and sub-components, which requires more data processing and checks.

Furthermore, expertise is also required to perform the matching of emission factors to the corresponding materials in consideration. It is usual that there are multiple data sources within a practitioner's database of emission factors for a single material. In order to ensure that the emission factors are matched fairly and appropriately to their respective materials, further efforts in the form of interviews and questionnaires are required to gather expert's

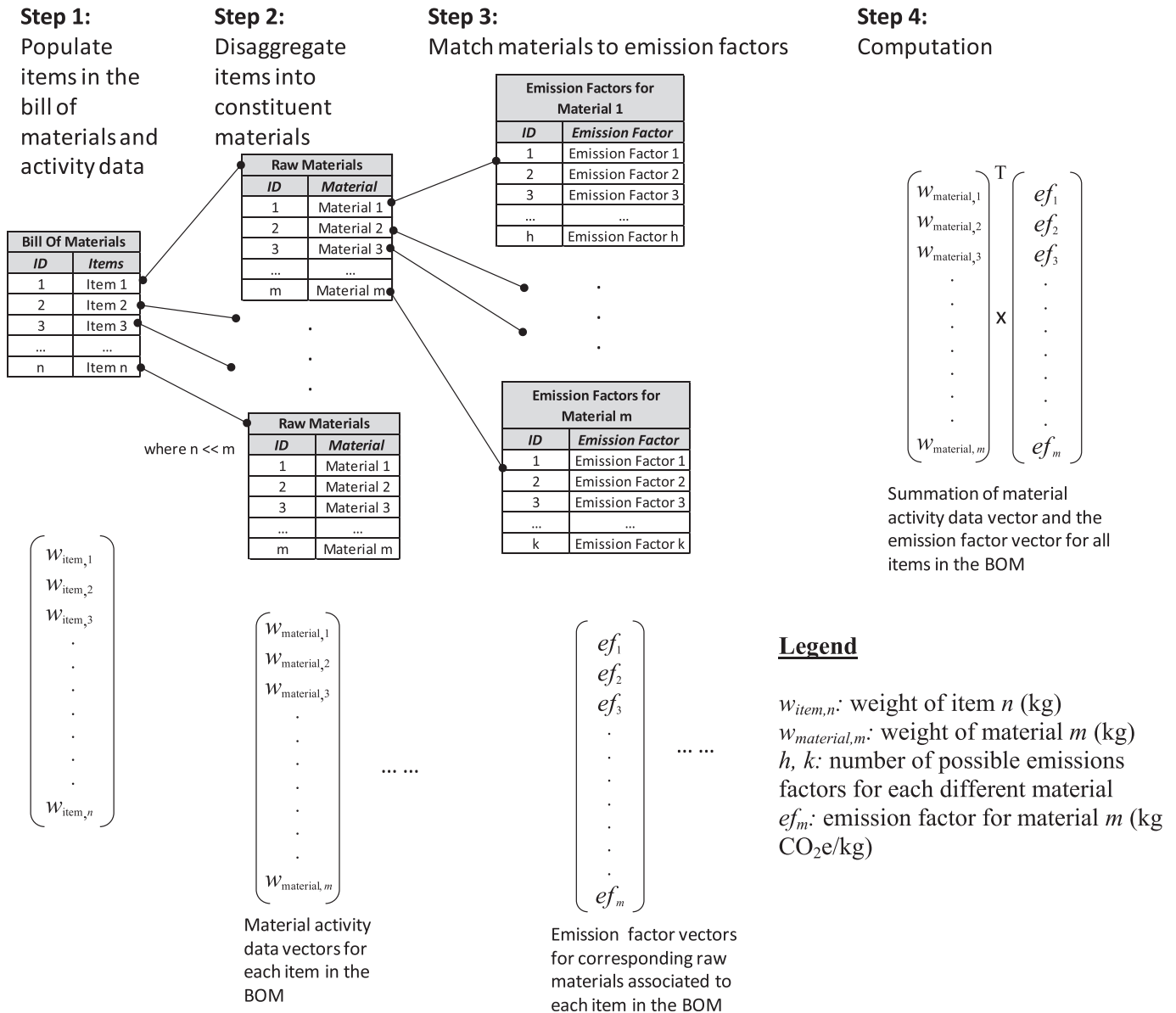


Fig. 1. Illustration of the current technique of embodied carbon footprint quantification.

domain knowledge. For instance, in this study's database, there exist over 70 possible values of emission factors for the embodied emissions of concrete. Details on the specifications of concrete such as its mix design and strength class needs to be utilized to determine the most representative emission factor for the material. The life cycle study by Gursel et al. (in press) has exhibited the impact of technical parameters on the greenhouse gas emissions of concrete.

2. Method

The purpose of this study is to propose a streamlined technique to efficiently quantify the embodied emissions for building materials. Following that, to validate the proposed technique, a comparison is made to illustrate the performance between the current technique and the proposed streamlined technique for quantification of the embodied carbon footprint of the building. The carbon footprint results of both techniques are compared, and the errors of the proposed technique are analyzed and discussed.

2.1. Current technique

The current technique of deriving the embodied carbon footprint of a building demands rigorous data collection and processing procedures. As illustrated in Fig. 1, the process of computing the embodied emissions of a building can be summarized into four steps. First, a completed BOM is required to be collected (Step 1). This step requires the collection of all detailed information regarding every item in the BOM, including the breakdown of the items into their raw material composition. Following that, items in

the BOM are then disaggregated into the constituent materials and the respective weights, also known as material activity data, are recorded (Step 2). This process creates a relationship between the list of raw materials and the items in the BOM. The next step (Step 3) involves the matching of the raw materials to the most appropriate emission factors. The practitioner selects the appropriate emission factors based on the data quality characteristics defined by the ISO/TS 14067 – time-related coverage, geographical coverage, technological coverage, precision, completeness, representativeness, consistency, reproducibility, sources of data and uncertainty of the information (ISO, 2013). The final step (Step 4) involves the aggregation of the material activity data and the associated emission factors to compute the final embodied carbon footprint due to the items in the BOM.

2.2. Proposed streamlined quantification technique

In contrast to the current traditional technique of deriving the embodied carbon footprint of all the items from the building's BOM, the proposed streamlined quantification technique aims to provide an estimate of the embodied carbon footprint as an early stage without a rigorous process. The purpose of this first cut estimate is to inform the practitioner on the order of magnitude of the final carbon footprint results to be expected and it also provides a starting point whereby data collection and refinement efforts can be further prioritized.

The proposed streamlined quantification technique is illustrated in Fig. 2. It comprises four steps. The first step (Step 1) involves the collection of the BOM of the building. However, the detailed

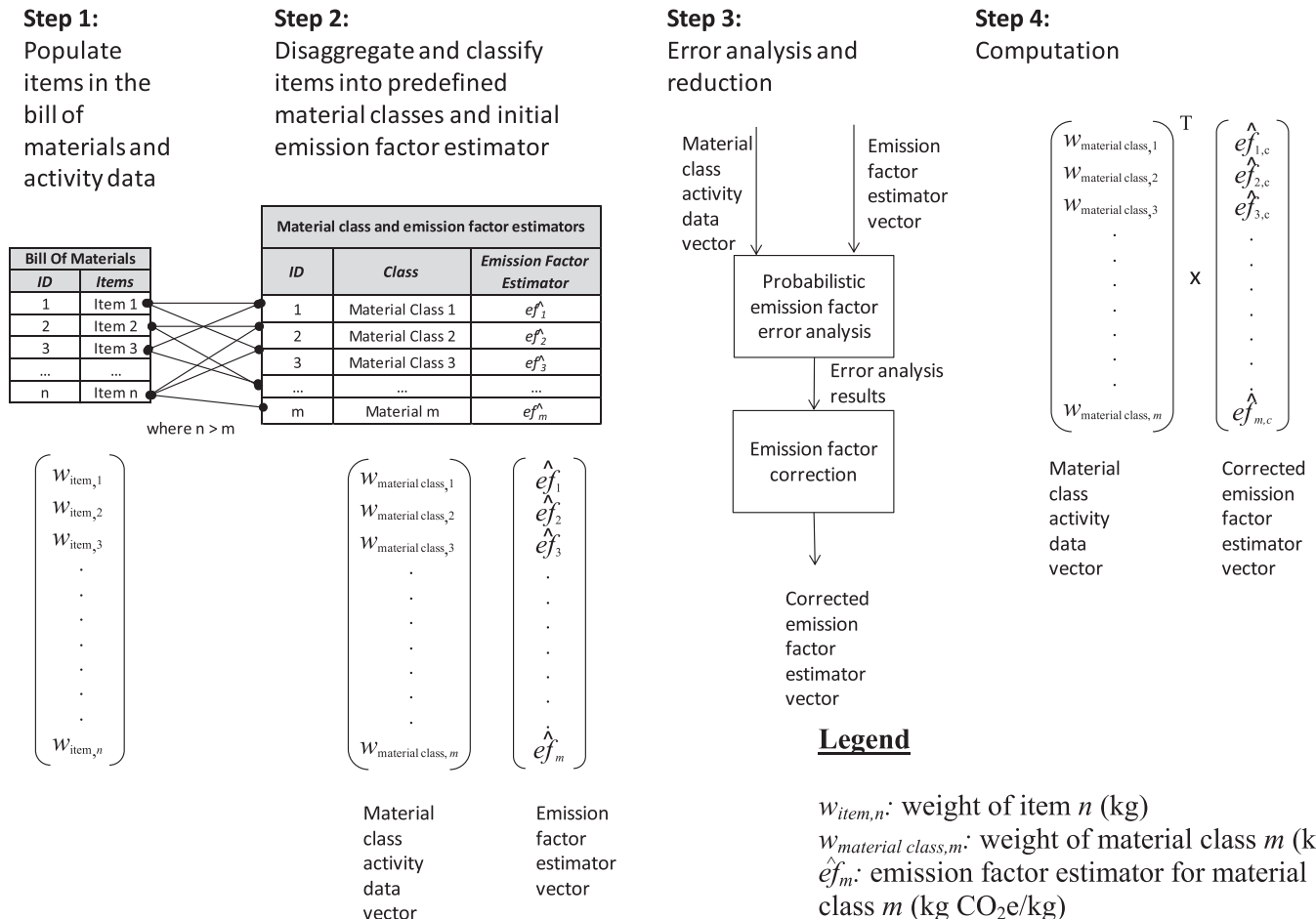


Fig. 2. Illustration of the proposed technique of embodied carbon footprint quantification.

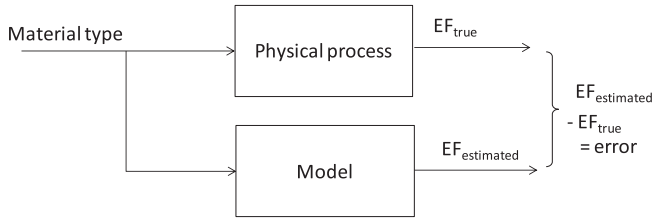


Fig. 3. Conceptual modeling of EF for a given material.

breakdown of the items into their composition may not be complete. Next (Step 2), the items in the BOM are disaggregated and labeled using one of the established *Material Class*. A *Material Class* is a generic category of materials without detailed specifications. For instance, stainless steel 306 is classified into the *Material Class* of stainless steel. Each *Material Class* consists of an emission factor estimator, together with its associated probability of error. More details on the probability of error are being discussed in subsequent sections. Additionally, at this step, a preliminary estimate can be established to estimate the order of magnitude of the expected embodied carbon footprint. Following that in Step 3, the estimation data is being utilized in a probabilistic model to predict the expected error for each data point in the computation. Each data point is assigned an expected error index. Based on the expected error index, the data points are classified into low, medium and high error bands based on their order of magnitude. Data points falling in the medium and high error bands are being reviewed and the emission factors for those data points are being manually revised. Finally in Step 4, the revised set of emission factor estimators and the material class activity data are combined to compute the embodied carbon footprint.

3. Theory and calculation

3.1. Emission factor estimator

The main objective of this work is to propose a streamlined technique for carbon footprint quantification of the building's embodied carbon emissions with limited data availability. It is acknowledged that when there is limited availability of data, the principle of completeness mentioned in the product carbon footprint standards (BSI, 2011; WRI, 2011; ISO, 2013) would be compromised. Furthermore, the principle of representativeness will also be compromised during the matching of the most appropriate emission factor to each material, especially when there

is a lack of localized emission factor data. Nevertheless, useful results can still be derived under the circumstances of limited data availability through the proposed the streamlined technique. This technique also aims to reduce the data processing requirements during the computation of the embodied emissions estimates, and guides the practitioner to improve results accuracy through targeted and selective data refinement.

In order to handle the effects caused by limited data availability, this technique approaches the problem by first overcoming the problem of emission factor matching to the material activity data. This can be done by providing a preliminary prediction of the emission factor for each individual material instead of attempting to manually match each material to its corresponding emission factor. This can be useful in frequently encountered scenarios whereby the material dataset at hand consists of minimal information, consisting of only the generic material types and amounts without further detailed specifications (e.g. strength class, metal alloy composition, type).

The preliminary prediction of the emission factor is based on a reasonable point estimate. Therefore, a reasonable point estimate has to be defined. For the purpose of this study, the reasonable point estimate of the emission factor is defined as the arithmetic mean of the entire population of emission factors that describes a particular material type. By taking the arithmetic mean of the population as the emission factor estimator, the sum of the deviations of all the emission factors in the population away from this arithmetic mean is zero. This can be shown by the following.

$$\bar{X} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \quad (1)$$

where:

$\bar{X}$ is the arithmetic mean

$x_1, x_2, x_3, \dots, x_n$ are the emission factors

n is the number of emission factor in a material class

Hence,

$$x_1 + x_2 + x_3 + \dots + x_n = \bar{X}n \quad (2)$$

The sum of the deviations of the individual emission factors from the arithmetic mean is calculated by the following.

$$\sum_{i=1}^n (x_i - \bar{X}) = (x_1 - \bar{X}) + (x_2 - \bar{X}) + (x_3 - \bar{X}) + \dots + (x_n - \bar{X}) \quad (3)$$

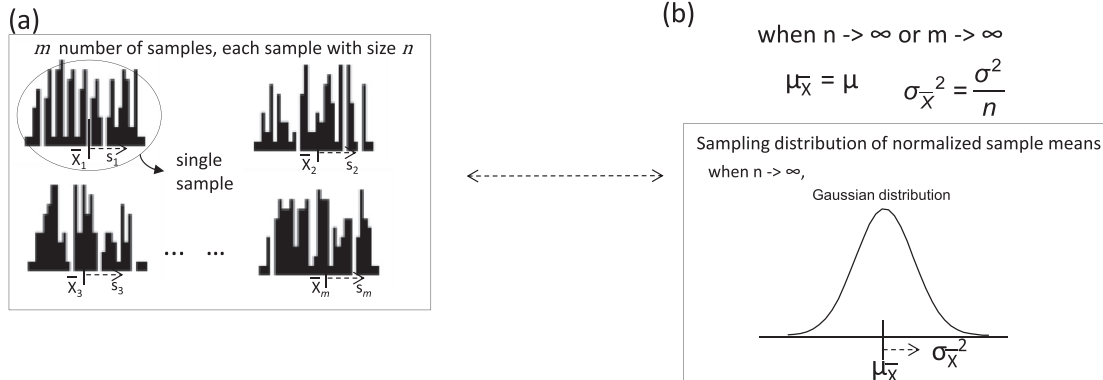


Fig. 4. Illustration of the central limit theorem applied in the context of emission factor estimation. (a) Left, shows the samples of emission factors (of a particular material class) from practitioners' databases. (b) Right, shows the sampling distribution of the sample means of emission factors obtained from practitioners' databases.

Table 1
Expected error index and corresponding error band.

Expected error index	Error band	Recommended action
Less than 0.01	Low	Data refinement not required
0.01–0.1	Medium	Data refinement required
0.1 and above	High	Data refinement required

Factorizing the above,

$$\sum_{i=1}^n (x_i - \bar{X}) = (x_1 + x_2 + x_3 + \dots + x_n) - \bar{X}n \quad (4)$$

Combining equations (2) and (4),

$$\sum_{i=1}^n (x_i - \bar{X}) = \bar{X}n - \bar{X}n = 0 \quad (5)$$

The mathematical property of the arithmetic mean implies that that the expected magnitude of underestimating and the expected magnitude of overestimating randomly selected emission factors from a set of emission factors are equal in the long run.

The conceptual modeling of the emission factor for any given material is illustrated in Fig. 3 below. For each material, due to the processes that are involved in manufacturing the material, greenhouse gases are released from the combustion of fossil fuels or direct release from chemical reactions. Therefore, a true emission factor (denoted as EF_{true}) exists for the manufacturing of the material. During the modeling of the material emission under limited information availability, the estimated emission factor (denoted as $EF_{\text{estimated}}$) is taken to be the average emission factor of the entire population of manufacturers for that material. For instance, if it is not possible to match a particular type of steel to its corresponding emission factor due to reasons such non-availability of precise data to match the specifications of the material; it will be assumed that a reasonable estimate for the steel in question will be the mean emission factor of the entire population of steel makers. It is noted that obtaining the arithmetic mean emission factor directly by gathering data from the entire population remains a formidable, if

Table 3
10 EF values of “Paint” collected from various data sources.

Data point	Original name	Data source ^a	EF value (kg CO ₂ e/kg material)
1	Paint, general	[1]	2.91
2	Single coat	[1]	0.437
3	Double coat	[1]	0.874
4	Triple coat	[1]	1.31
5	Waterborne paint	[1]	2.54
6	Solventborne paint	[1]	3.76
7	Acrylic varnish, 87.5% in H ₂ O, at plant/RER S	[2]	1.75
8	Alkyd paint, white, 60% in H ₂ O, at plant/RER S	[2]	2.46
9	Alkyd paint, white, 60% in solvent, at plant/RER S	[2]	2.51
10	Paint ETH S	[3]	1.79

^a [1] Inventory of Carbon & Energy (ICE), [2] Ecoinvent, [3] ETH-ESU 96.

not impossible task. Therefore, a way to reliably approximate such a conceptual model has to be established.

3.2. Derivation of the emission factor estimator

The Central Limit Theorem (CLT) enables the inference of an initially unknown mean of a population by taking random samples from the population. The CLT states that when random samples (of size n) are drawn from a population of any distribution type with a finite mean (μ) and variance (σ), the sampling distribution derived from the means generated by all the random samples will approach a normal distribution as n approaches infinity. Additionally, the mean of the sampling distribution ($\mu_{\bar{X}}$) will be equivalent to the population mean ($\mu_{\bar{X}} = \mu$) and a corresponding variance of ($\sigma_{\bar{X}} = \sigma^2/n$) when the sample size approaches infinity or the number of samples taken approaches infinity (Montgomery and Runger, 2011; DeCoursey, 2003). The mathematical property of CLT enables the estimation of the population mean from a single random sample to a certain degree of accuracy. In addition, the CLT provides a mathematical framework to estimate the accuracy of the estimators to a specified level of accuracy.

Table 2
Emission factor estimators.

Material class	Data sample size (N)	Mean emission factor, kg CO ₂ e/kg material ($\bar{X}$) ^b	Sample standard deviation, kg CO ₂ e/kg material (s)	Tolerance, kg CO ₂ e/kg material ($\bar{X}\tau$, where $\tau = 5\%$)	P(error)	Confidence level	Data source ^a
Aluminum	31	7.47	4.784	0.374	0.664	0.336	[1, 2, 3, 5, 6]
Asphalt	5	0.08	0.008	0.004	0.343	0.657	[5]
Cement	26	0.61	0.227	0.030	0.502	0.498	[2, 5]
Ceramics	8	1.01	0.667	0.050	0.837	0.163	[2, 3, 5, 6]
Clay	7	0.40	0.131	0.020	0.700	0.300	[2, 5]
Concrete	79	0.12	0.045	0.006	0.241	0.759	[2, 3, 5, 6, 11]
Copper	4	2.86	1.471	0.143	0.858	0.142	[5, 9]
Glass	11	1.33	0.622	0.067	0.729	0.271	[2, 3, 5]
Mortar	10	0.25	0.160	0.013	0.808	0.192	[5, 6, 12]
Other metals	11	3.68	3.865	0.184	0.878	0.122	[5]
Paint	10	2.03	0.998	0.102	0.755	0.245	[2, 5]
Plaster	6	0.31	0.254	0.015	0.887	0.113	[2, 5]
Plastics	23	3.95	1.853	0.198	0.614	0.386	[5, 10]
Rubber	13	2.68	1.053	0.134	0.655	0.345	[2, 3, 4, 5, 6, 7, 9]
Sand	7	0.05	0.113	0.003	0.951	0.049	[3, 5, 9, 11]
Sealants & adhesives	11	5.79	4.103	0.290	0.820	0.180	[2, 5, 10]
Stainless steel	10	4.80	0.876	0.240	0.409	0.591	[8, 9]
Steel	55	1.83	0.837	0.092	0.417	0.593	[5]
Stone, gravel	12	0.11	0.196	0.006	0.922	0.078	[2, 5]
Wood	10	0.85	0.175	0.042	0.464	0.536	[5]

^a [1] BUWAL, [2] Ecoinvent, [3] ETH-ESU 96, [4] Franklin USA 98, [5] Inventory of Carbon & Energy (ICE), [6] IDEMAT 2001, [7] Industry Data 2.0, [8] International Stainless Steel Forum, [9] PE International, [10] Plastics Europe, [11] SIMTech, [12] World Steel Association.

^b Values are rounded to the nearest 2 decimal point.

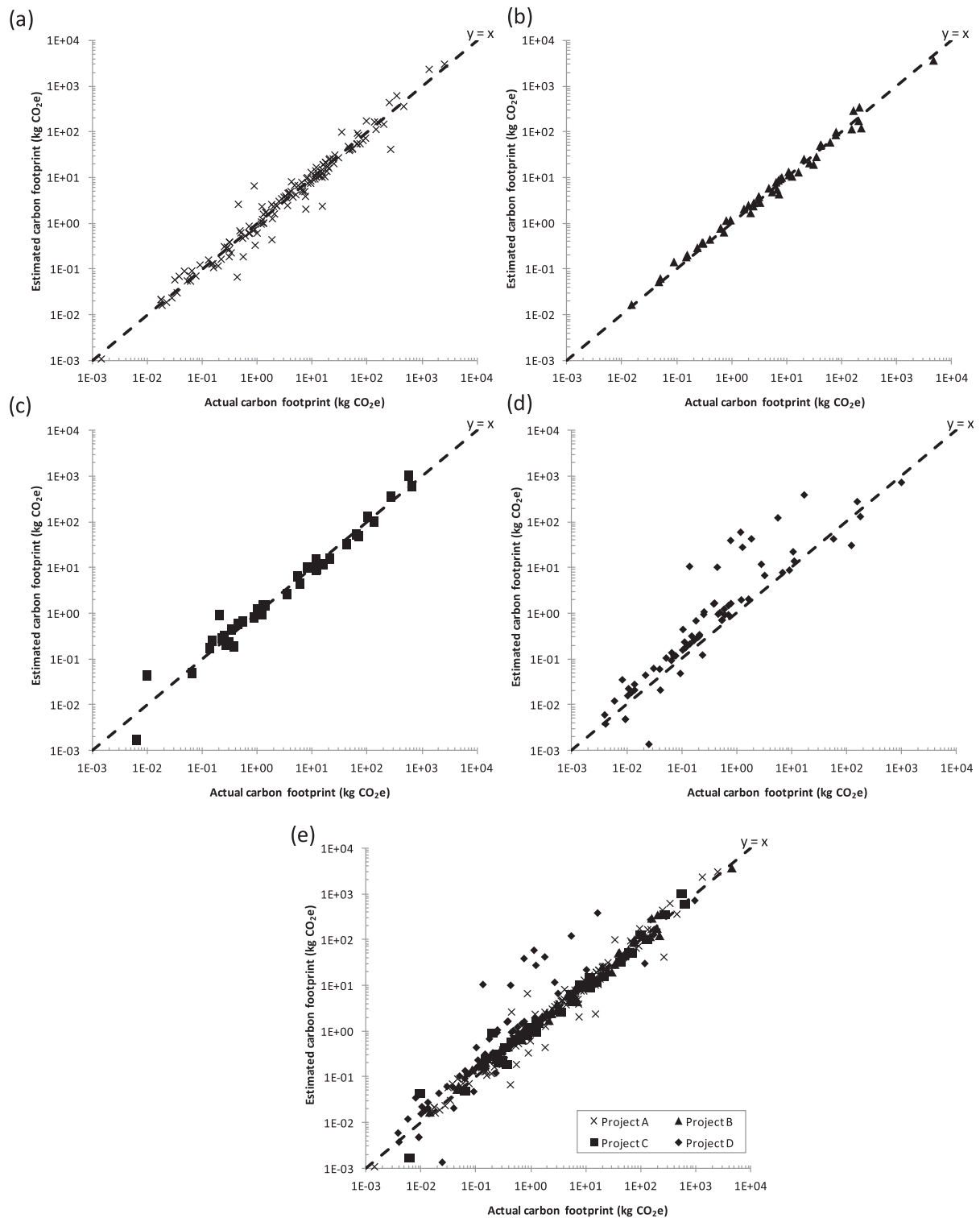


Fig. 5. Graphs of estimated carbon footprint versus actual carbon footprint of the four projects.

Fig. 4 shows the application of the CLT in the context of establishing the emission factor estimator. Firstly, the population distribution of emission factors for a particular material production activity is assumed to have a finite mean (μ) and population variance (σ). However, these values (μ and σ) are impractical to be obtained as it requires the measurement of the entire population. Nevertheless, very often within a practitioner's database collection

of emission factors, there exist population samples of the emission factors for diverse types of materials. These samples for each material class can be gathered easily and computed (illustrated in Fig. 4a). When the number of samples approaches infinity, the sample variance approaches the population variance. Based on the CLT, the mean derived from the sampling distribution of the sample means will be equivalent to the population mean (which is defined

Table 4

Correlation coefficient of the estimated carbon footprint and the actual carbon footprint for the four projects.

Project	Correlation coefficient
A	0.978
B	0.998
C	0.941
D	0.868
Overall	0.981

Table 5

Comparison of initial estimates of embodied carbon footprint and the actual embodied carbon footprint.

Project	Estimated embodied carbon footprint ^a	Actual embodied carbon footprint ^a	Error (%)
A	126.48	100	+26%
B	86.83	100	−13%
C	120	100	+20%
D	125	100	+25%

^a The actual embodied carbon footprint is normalized to a base index of 100 for all projects. The estimated carbon footprint is normalized with reference to the actual embodied emission.

Table 6

Number of data points in error band.

Project	Number of data points in error band		
	Low	Medium	High
A	127	15	2
B	45	9	3
C	30	4	3
D	58	8	3
Overall	260	36	11

as the emission factor estimator), and the distribution approaches a Gaussian distribution, which is the basis to derive the probability of the sample mean being a predefined expected distance away from the population mean (illustrated in Fig. 4b).

3.3. Error estimation

The relationship that links the sample mean to the population mean enables the computation of the probability of the population

mean to be within a predefined range. Conversely, if a range is initially predefined, the probability of the population mean to lie within that predefined range can also be determined (also known as the confidence level). The distance away from the mean normalized by the standard deviation of the sampling distribution can be measured by equation (6).

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \quad (6)$$

With the calculated normalized distance (t), the probability of the sample mean ($\bar{X}$) that will lie within the margin of error, $\pm(\bar{X} - \mu)$, away from the population mean (μ) can be estimated with the normal distribution for known population variance, or from the t distribution for unknown population variance, assuming that the sample size is sufficiently large or the sample is drawn from a population that is approximately normal. By setting a predefined level of tolerance (τ) and multiplying it to the sample mean $\bar{X}$, we can define the expected magnitude of the variation of the sample mean away from the population mean at the specified level of tolerance. Equating $(\bar{X} - \mu)$ to $\bar{X}\tau$,

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} = \frac{\bar{X}\tau}{\frac{s}{\sqrt{n}}} \quad (7)$$

where:

$$0\% < \tau < 100\%$$

By specifying the level of τ , the level of tolerable expected variation away from the mean can be defined. Correspondingly, the probability that is associated to t can be interpreted as the confidence level that the population mean will be away from the sample mean with a magnitude of $\bar{X}\tau$. Taking the complement of this confidence level, the probability of an error that would occur, $P(\text{error})$, can be estimated (probability that the distance between the population mean (μ) and the sample mean ($\bar{X}$) will be more than $\bar{X}\tau$). The value of $P(\text{error})$ provides a measure to the expectation of which the emission factor estimator will not estimate the true mean emission factor to within the predefined margin of error. Hence, the value of $P(\text{error})$ can be further utilized to provide an estimate for the expected relative error when using the emission

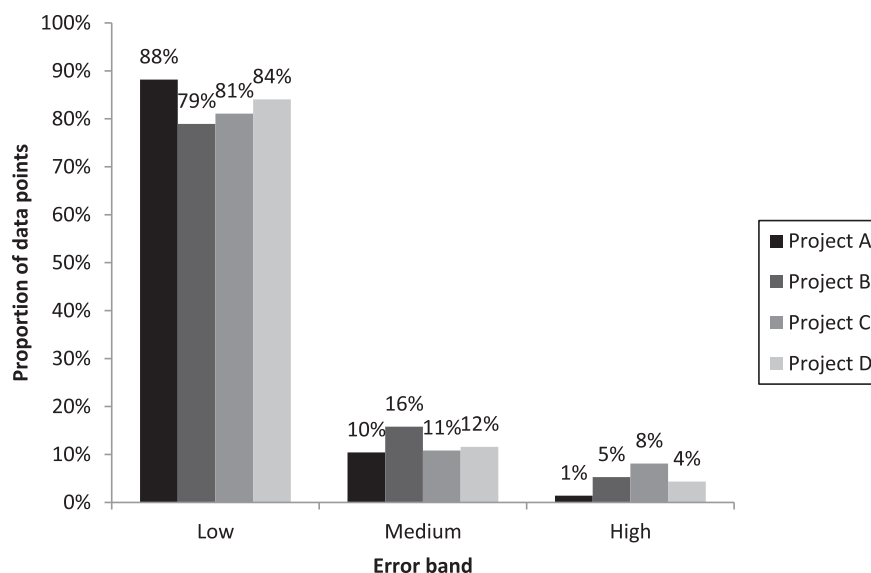


Fig. 6. Proportion of the data in the three error bands.

Table 7
Proportion of data points in the respective error bands.

Project	Proportion of data points in error bands ^a			
	Low	Medium	High	Medium and high
A	88%	10%	1%	12%
B	79%	16%	5%	21%
C	81%	11%	8%	19%
D	84%	12%	4%	16%
Overall	85%	12%	4%	15%

^a Rounded to the nearest whole number.

Table 8
Final estimated embodied carbon footprint versus actual embodied carbon footprint after error analysis and reduction.

Project	Estimated embodied carbon footprint ^a	Actual embodied carbon footprint ^a	Error (%)
A	99.61	100.00	−0.39%
B	99.75	100.00	−0.25%
C	99.85	100.00	−1.15%
D	103.65	100.00	+3.65%

^a The actual embodied carbon footprint is normalized to a base index of 100 for all projects. The estimated carbon footprint is normalized with reference to the actual embodied emission.

factor estimators in the computation of the carbon footprint of each item.

Two indicators for the measurement of error are defined – expected error index and actual absolute relative error. The expected error index represents estimation for the expected variation away from the mean value of the results for each item in the BOM. The expected error is proportional to the weight of the object, the spread of the emission factor estimator and the probability of error of the estimator. It is defined as follows:

$$\text{Expected error index}_i = \frac{\sum_{j=1}^M w_{ij} \times s_{ij} \times P(\text{error})_{ij}}{\sum_{k=1}^N \sum_{l=1}^M w_{kl} \times s_{kl} \times P(\text{error})_{kl}} \quad (8)$$

Table 9
Accuracy of classification of errors.

Accurate classification	“Over” classification	“Under” classification
93.81%	4.89%	1.30%

where:

w_{ij} is the weight of the j th material class for the i th item in the BOM

s_{ij} is the sample standard deviation of the j th material class for the i th item in the BOM

$P(\text{error})$ is the expected probability of error of the j th material class for the i th item in the BOM

w_{kl} is the weight of the k th material class for the l th item in the BOM

s_{kl} is the sample standard deviation of the k th material class for the l th item in the BOM

N is the total number of items in the BOM

M is the total number of material class

The actual absolute relative error provides a similar measure of error, taking the difference between the estimated values and the actual values of the carbon footprint as a basis. The purpose of the actual absolute relative error is to provide a benchmark to compare the predictive performance of the error index to the actual error that occurs in reality. It is defined as follows:

$$\text{Actual absolute relative error}_i = \frac{|\widehat{CF}_i - CF_i|}{\sum_{j=1}^n CF_j} \quad (9)$$

where:

$\widehat{CF}_i$ represents estimated carbon footprint of the i th item in the BOM

CF_i represents actual carbon footprint of the i th item in the BOM

CF_j represents actual carbon footprint the j th item in the BOM

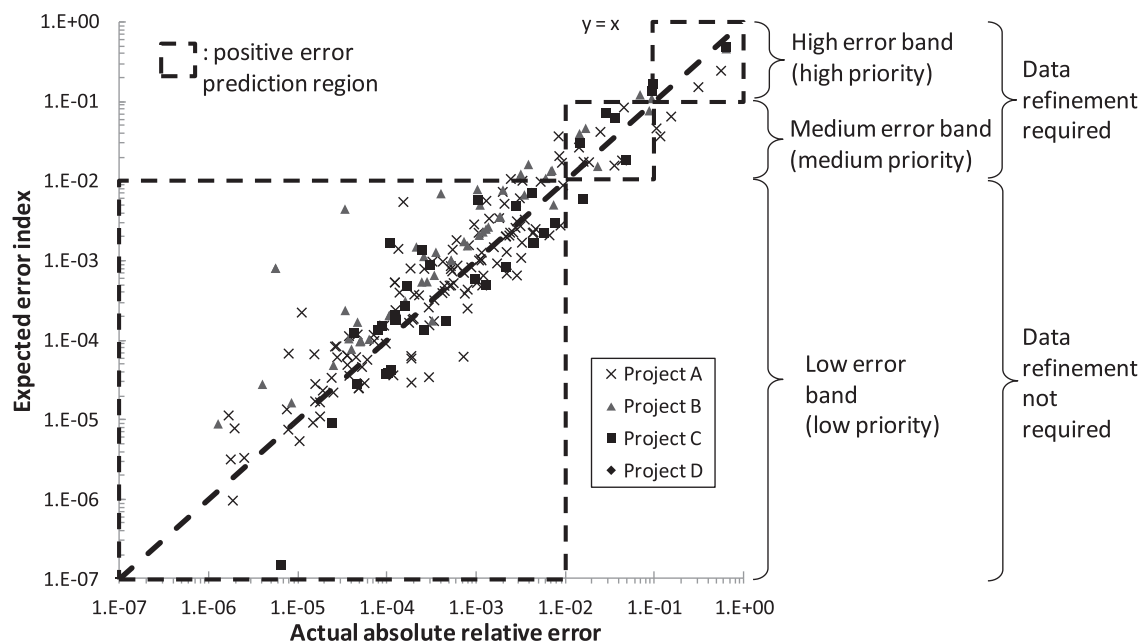


Fig. 7. Graph of expected error index versus actual absolute relative error.

n represents the total number of items in the BOM

In reality, only the expected relative error can be calculated to forecast errors, while the actual absolute relative error is calculated retrospectively for the purpose of comparison in this paper.

Each item in the BOM can be subsequently assigned to an error band based on the expected error index calculated from above. The error bands are classified based on the order of magnitude as shown in Table 1.

The error band serves as an indicator to prioritize the efforts to make manual corrections to the emission factors for the items. For items that fall within the “low” error band, it indicates that the predicted error is low and the estimated carbon footprint is reasonable for those items. On the other hand, items that are classified as “high” error band indicates that those items need to be further looked into and a more appropriate emission factor have to be manually selected. For those items in the “medium” error band, it is recommended that these items to be further examined after the “high” error band items are addressed. Items which are classified as “low” error band are expected to have low errors in estimation and hence no further action is required to correct the carbon footprint estimation by the practitioner.

4. Results

4.1. Case studies background

Four construction projects – referred to as Project A, Project B, Project C and Project D – are used as case studies for the streamline quantification method to estimate their embodied carbon emissions from the BOM data consisting of a total of 307 individual items. Project A involves the construction of a shopping mall, Project B is the construction of a commercial building, Project C is the construction of a bridge and Project D is the construction of a research laboratory. These four projects have also undergone the rigorous current method of carbon footprint quantification. In order to make a comparison, the results of the embodied carbon emission done manually with the traditional method are used as the benchmark. Results derived from the traditional technique are also labeled as actual carbon footprint whereas the results derived from the proposed technique are labeled as estimated carbon footprint. Therefore, the error in the results is the difference between the proposed streamlined quantification technique and the traditional method of quantification.

4.2. Emission factor estimators

Applying the concept of the proposed technique, a table of emission factor estimators is prepared accordingly. This table of emission factor estimators is established as a proof-of-concept and is subsequently used in the case study for carbon footprint estimation. For the purpose of this study, the parameter for the tolerance (τ) for the variation of the estimator is set at 5%. The determination of the optimal tolerance value is not within the scope of this study and will be discussed in subsequent works. Nevertheless, if the tolerance is set too low, the confidence level of the emission factor estimators will in turn be too small, whereas in the case whereby the tolerance is set too high, the confidence level will be too large to make any meaningful conclusions.

To establish the table of emission factor estimators, data is collated from various life cycle inventory and carbon footprint databases. They include: Inventory of Carbon & Energy (Sustainable Energy Research Team (SERT), 2011), Ecoinvent (Centre for Life Cycle Inventories, 2010), IDEMAT 2001 (Delft University of Technology, 2008), ETH-ESU 96 (ESU-services, 1996), BUWAL

(Swiss Packaging Institute, 2000), World Steel Association (World Steel Association, 2000), PE International (PE International, 2010), Franklin USA 98 (Franklin USA 98, 1998), Plastics Europe (Plastics Europe, 2005), International Stainless Steel Forum (International Stainless Steel Forum, 2005) and also carbon footprint studies conducted by SIMTech (SIMTech, 2010). While most of the underlying data in this current set of prototype emission factor estimators are from the European region, it has provided a reasonable estimate as exhibited in the following sections. Therefore, by including a higher proportion of localized data source, the resulting emission factor estimation range is expected to be narrower.

As these data sources may vary in the format of storing carbon footprint information, preprocessing steps have to be done in the compilation of the carbon footprint emission factors. For data sources that presented a final emission factor for the materials (e.g. kg CO₂e/kg material), the figures are taken directly from these data sources. For consistency, quantities that are not represented in SI unit are converted into SI unit representation (e.g. g CO₂e will be converted to kg CO₂e). For data sources that presented individual greenhouse gas emission quantities associated to the production of the materials, the individual greenhouse gas emission quantities will be converted into kg CO₂e and aggregated up to obtain the emission factor for each material.

The emission factors from these data sources are then systematically categorized into their respective material classes. Following that, the mean emission factors are computed for each material class. The mean emission factors will be used to perform the initial estimation of the carbon footprint from the bill-of-material. In addition, the sample standard deviation, tolerance, confidence level and the associated probability of error, $P(\text{error})$, are computed. The values of $P(\text{error})$ will be utilized in the subsequent error prediction and correction steps after the initial estimation is made. Table 2 forms the basis of emission factor estimation used in the following case studies.

To further illustrate the construction of Table 2, the case for “Paint” will be discussed. The same method will be used to process all the other material classes. Table 3 shows the raw data that are collated from the respective databases.

Using equation (1), the arithmetic mean ($\bar{X}$) for the “Paint” material class is calculated as 2.03 kg CO₂e/kg material. Correspondingly, the sample standard deviation for the set of EF values is 0.998. By setting tolerance level (τ) to 5%, the tolerance ($\bar{X}\tau$) is 0.102. Using equation (7), the normalized distance away from the mean at the predefined tolerance level (t) is calculated as 0.322. With the tolerance level calculated, the corresponding probability of error, $P(\text{error})$, can be derived from a standard 2-tailed t-distribution function, degree of freedom = 9, in which this case is 0.755. The level of confidence can be calculated using the complement of $P(\text{error})$, which is 0.245.

4.3. Initial estimation and error analysis

Utilizing the material classes defined in Table 2, the BOM of each project is examined and each item is assigned the most appropriate material class that describes the item. Once the material classification is performed, the corresponding weight information of each material class is computed. The weight information is then combined with the mean emission factor for the respective material classes to estimate the carbon footprint associated to each material class.

The initial results computed from the proposed streamlined technique are compared to the results derived from the current method for all the four projects. The above procedure yielded 614 results – half of which from the proposed streamlined technique, the other half derived from the traditional technique of quantification. The data of the estimated carbon footprint of all the individual BOM items are plotted against the corresponding carbon footprint

results obtained from the traditional technique shown in Fig. 5 below. The graph illustrates an approximately linear relationship between the estimated carbon footprint and the actual carbon footprint derived from the traditional technique. The correlation coefficient associated to the data points is 0.981 (Table 4). Individually, the correlation coefficients for data are 0.978 for Project A (Fig. 5a), 0.998 for Project B (Fig. 5b), 0.941 for Project C (Fig. 5c) and 0.868 for Project D (Fig. 5d). This indicates that the estimated carbon footprint derived from the proposed technique serves as a reasonable predictive measure of the actual carbon footprint.

The magnitudes of the total embodied carbon footprint were then aggregated for all the four projects using the results derived from the two techniques. Further examination on the magnitude of the total embodied carbon footprint indicate that the initial estimation using the mean emission factor as the estimator resulted in the overestimation in the embodied carbon footprint for all the four projects. The estimation errors range from +15% to +26% (shown in Table 5). While the results produced by the proposed technique at this stage still remains largely imprecise to estimate the total embodied carbon footprint, the results can be further utilize to determine the relative carbon footprint of each item in the BOM and this information is used for further analysis for error reduction.

To yield more accurate results, the expected error index is computed for each item in the BOM using equation (8). Using the expected error index, each individual item in the BOM is given a low, medium or high error band as defined in Table 1. The number of data points in each error band is given in Table 6. Correspondingly, the distribution of the data in the error bands for the four projects are shown in Fig. 6. It can be observed that majority of the items in the BOM are classified in the low error band (between 79% and 88% for each project). Between 10 and 16% of the data were classified in the medium error band, while between 1 and 8% were classified in the low error band. The error distribution shows that majority of the items are expected to have low level of estimation error. Furthermore, the results also show that when higher magnitudes of error occur, it is originates from minority of the items. This observation will be further discussed in the subsequent section.

4.4. Error reduction and final results

Continuing from the error banding results, the BOM items that were classified under the high and medium error band were manually evaluated and their respective emission factors were refined in the same manner as the traditional technique. However, it is noted that while the traditional technique requires all items of the BOM to undergo rigorous emission factor selection and matching, the proposed streamlined technique only applies the rigorous procedure only to items identified with high and medium errors, which forms between 12 and 21% of the total data points in the four projects (Table 7). This also implies that the use of the proposed estimator is able to estimate between 79% and 88% of the data points with a low level of error. This is a useful result to guide practitioners to focus their data collection efforts only to those data points which are expected to cause higher levels of error in the estimates, while majority of the data points can be estimated with a high level of accuracy.

Table 8 shows the results of the final estimated embodied carbon footprint after the data points that are classified as high and medium errors were manually evaluated and the emissions factor corrected by the practitioner. The final estimated embodied carbon footprints for the four projects were then compared to the set of results that were originally produced using the traditional technique of carbon footprint assessment. The results showed that the error of estimation range from −0.25% to +3.65%.

A detailed analysis of the errors is shown in Fig. 7. The graph shows a high correlation between the expected error index and the

actual absolute relative error, with an overall correlation coefficient of 0.92. This implies that with the calculated expected error index, the actual absolute error can be predicted to a reasonable level of accuracy.

The graph is further demarcated to show the three error bands. Majority of the data points falling in the low error band (expected error index less than 0.01). These data points are expected to have a low degree of estimation errors and hence, data refinement is not required.

Data points with expected error index of more than 0.01 are required to be further refined as the carbon footprint emissions derived from the default estimators are deemed to have a high level of uncertainty. As these data points requires further effort in examination and revision by the practitioner, the error bands will assist the practitioner to prioritize the efforts in the treatment of the data points. The data points which are in the high error band (expected error index of more than 0.1) should receive attention first, followed by those that lie in the medium error band (expected error index of between 0.01 and 0.1). Within each band, the magnitude of the expected error index can also help rank the data points. Data points with a higher expected error index should be revised first. This assists the practitioner to revise data points starting with the one with the highest expected level of error.

5. Discussion

5.1. Performance of the expected error index and error band classification

The expected error index is a calculated normalized index value that is a function of: (1) the probability of error of the emission factor estimator with a specified level of τ (5% in this study), (2) the sample standard deviation of the emission factor estimator, and, (3) the material class activity data (Equation (8)).

The probability of error of the emission factor estimator is derived based on the assumption that the sample of emission factor is drawn from an approximately normally distributed population of emission factors for the material class. If the sample size is sufficiently large, the assumption of normality becomes secondary. This assumption enables the computation of the probability that the sample is unable to provide a good estimate of the true population mean within a predefined level of tolerance. This serves as a property to describe the uncertainty of the emission factor estimator. The sample standard deviation of the emission factor estimator is also taken into consideration to compute the expected error index as it is assumed that the larger the spread of the sample data, the higher the level of uncertainty in the estimation. Finally, as the material class activity data itself is also part of the computation of the carbon footprint, it has to be taken into account when computing a measure for the expected error.

The three parameters were combined and normalized to sum up to 100% for all the data points (Equation (8)) and was found to be highly correlated to the actual errors. Hence, the expected error index was used to classify the data points into error bands.

Table 9 shows the performance of the classification based on the 307 data points from the four projects in this study. Accurate classification is defined as the predicted error band is equivalent to the actual error band. “Over” classification is defined as the predicted error band of a higher error band compared to the actual error band. For instance, if the actual error band is low while the predicted error band is medium, it is considered as “over” classification. Conversely, “under” classification is defined as the

predicted band under predicting the actual error band. For instance, if the actual error band is medium while the predicted error is low, it is considered as “under” classification.

Based on the data points from the case study, the expected error index achieved an accurate classification of 93.81%, 4.89% of the data were being over-classified while 1.30% of the data were being under-classified. This largely accurate classification of errors contributes to the final revised embodied carbon footprint produced by the proposed streamlined technique to be only a small variation away from the actual embodied carbon footprint computed by the current technique. Only with accurate determination of the errors in estimation can the results be corrected towards getting a more accurate value. Without the knowledge of the errors, practitioners will have to refine all the data points to derive the final carbon footprint. This brute force approach will negate any time savings obtained from using the emission factor estimators. Hence, this technique relies on an accurate error estimation step to establish an accurate final result.

5.2. Use of mean and median as estimator

As discussed in Section 3.1, the use of mean as the estimators is due to the property that the expected sum of the deviation of the emission factors away from the mean is zero. This implies that the expected magnitude of under or overestimation is zero over the long run. However, extra care has to be taken when the sample consists of extreme outliers. The presence of outliers may skew the value of the mean, which may affect the accuracy of estimation. Therefore, in the case of the presence of outliers, outliers may be considered to be excluded from the sample of emission factors.

The use of median was also explored during the development of the streamlined technique. The advantage of utilizing the median as the measure of central tendency is that it is less affected by outliers as opposed to the mean. The median is also defined as the middle score of a set of data, which separates the data into two equal proportions of which one group will be above the median value and the other group will be below the median value. This implies that there will be equal probability of underestimating and overestimating the true emission factor if the median is being used as the estimator.

Therefore, if the objective is to minimize the magnitude of under or overestimating the emission factor, the mean should be used as the estimator. If the objective is to minimize the probability of under or overestimating the emission factor, the median should be used as the estimator. In the proposed technique, the mean is used as the estimator as the primary objective is to minimize the expected magnitude of under or overestimation derived from the technique.

5.3. Limitations

The efficacy of the proposed streamlined technique to estimate the embodied carbon footprint is determined by the adherence of the data points to the assumptions of the CLT. The data samples are assumed to be drawn randomly from a population. Therefore, to achieve desired performance, the data samples have to be truly random sample from a population of interest. It is recommended that the underlying data used to compute the emission factor estimators be drawn from a diverse collection of databases. In this study, a mix of different available databases was utilized in establishing the emission factor estimators.

Another limitation of the technique lies in the error estimation procedure as it is not possible to discern if the estimated emission

factor is expected to be an over estimate or an under estimate. This is largely due to the underlying definition of a confidence interval that this technique is based upon. A confidence interval provides the range in which the true population mean is expected to lie within at a specified level of confidence. However, no information can be derived from the confidence interval if the sample mean is an over estimate or an under estimate of the true population mean. Nevertheless, through the case studies, the magnitude of expected error index provides a close approximation of the actual relative magnitude of error, which can be utilized to assist in the correction of errors.

5.4. Future work

The proposed streamlined technique can potentially be part of the foundational work towards an automated procedure for assessing the embodied carbon footprint from a large set of BOM. With the use of the emission factor estimators and the error reduction procedure, the embodied carbon footprints of buildings were obtained to a high level of accuracy as exemplified by the four case studies. The technique also minimized the requirement of human inputs in determining the emission factors for majority of the items in the BOM, with the exception of items with higher levels of uncertainty and errors. Further work can be done to improve the accuracy of the emission factor estimators and error reduction procedure.

Currently, the emission factor estimators and material classes were being determined manually. Therefore, in order to move towards automatic carbon footprint computation, a technique to automatically prepare the emission factor estimator table has to be established. Firstly, the interface with different databases and the accurate parsing of data must be performed in order to prepare the data derived from various database sources into a common format for processing. Following that, a combination of classification and clustering techniques can be applied to the attributes related to each emission factor. The classification can be carried out to autonomously label each emission factor with the appropriate material class. Clustering can also be done to discover potential new material class previously not in the existing list. Therefore, with the appropriate use of data mining techniques, the emission factor estimators may potentially be established autonomously.

Similar to emission factor estimators and material classes, the material activity data were being determined manually in this study. Potentially, the classification of activity data can also be done autonomously. Currently, the material activity data is established based on the text content from the BOM, which may be in an unstructured or semi-structure format. Therefore, text mining can be explored to enable the categorical labeling of each piece of material activity data based on textual content from the BOM.

In combination, it is envisaged that the automatic establishment of the table of estimators, labeling of material activity data directly from the BOM and the proposed streamlined quantification technique will enable the establishment of a highly autonomous system in determining the embodied carbon footprint of items derived from a large set of BOM, requiring minimal intervention from practitioners' performing embodied carbon footprint quantification.

6. Conclusion

Buildings consume a significant proportion of the world's resources, mostly in the form of materials usage. In addition, with the ongoing efforts focusing on reducing the operational energy

consumption of buildings, the embodied emissions of buildings are expected to increase in proportion in the future. With the increasing focus placed on the embodied emissions of buildings, organizations in the construction industry will face requirements to report information regarding the embodied emissions of buildings. However, in order to obtain the embodied emission of buildings, tremendous efforts have to be placed as the process is data intensive. In order to overcome this challenge, a streamlined technique is proposed to minimize the efforts required by practitioner to obtain the embodied carbon footprint. Based on the four projects presented as case study for the proposed technique, the practitioner would be required to manually match the appropriate emission factors to the activity data for between 12 and 21% of the data points. This is in contrast to the current traditional technique, which requires the manual matching of 100% of data points. Based on four case studies, the resultant deviation of the computed embodied carbon footprint from the proposed technique and the current technique was between -0.25% and $+3.65\%$. The proposed technique can also form part of the foundational work to move towards the automation of the process of computing embodied carbon footprint. Future work utilizing techniques derived from data mining is also proposed to assist in the automatic categorization of data points associated to the material activity data and the emission factors which was currently performed manually.

Appendix 1

Table A
Representative sample of 30 items from Project A.

No.	Item	Total weight (kg)	Material composition ^a
1	Concrete, in-situ	25,972,087.22	100% concrete
2	Rebar	1,301,600.00	100% steel
3	Concrete block	472,044.45	100% concrete
4	Concrete precast	117,600.00	100% concrete
5	H-beam	97,000.00	100% steel
6	Reinforced concrete piles	56,332.80	1% steel, 99% concrete
7	Marble	35,324.32	100% stone
8	Aluminum, curtainwall	15,492.00	100% aluminum
9	Glass	12,825.00	100% glass
10	Steel pipe	10,711.50	100% steel
11	Aluminum/trellis	9968.40	100% aluminum
12	Moisture resistant fibrous plasterboard ceiling	6963.60	97% plaster, 3% fiberglass
13	Glass block	5175.00	100% glass
14	Angle cable trays	3745.32	100% galvanized Steel
15	Heavy duty stainless steel angle	2732.22	100% stainless steel
16	Drainage cell with geofabric	2100.60	100% plastic
17	PU elastomeric seamless membrane	1397.70	100% polyurethane
18	Steel pipes	992.00	100% steel pipe
19	ELV cable	800.00	80% copper, 20% PVC
20	Water closet	560.00	100% ceramic sanitary product
21	Ductile iron pipe	342.00	100% iron
22	Aluminum ventilation louver	220.00	100% aluminum
23	Access cover	150.00	100% iron
24	Irrigation system – pump	100.00	100% PVC
25	Mirror	90.00	99% glass, 1% aluminum
26	Paint	54.24	100% paint
27	Hosereel electrical pump	33.80	70% steel, 30% copper
28	Toilet roll holder	15.00	100% polyethylene
29	Robe hook	5.00	100% stainless steel
30	Air release valve	0.30	100% brass

^a Note: the material composition is provided “as-is”, received from the quantity surveyor.

Illustration of computation of carbon footprint using current technique

Table B
Summary of computation using current technique.

No.	Item	Total weight (kg) [A]	Weighted average emission factor (kg CO ₂ e/kg) [B]	Emissions (kg CO ₂ e) [A × B]
1	Concrete, in-situ	25,972,087.22	0.0862	2,238,793.92
2	Rebar	1,301,600.00	1.01	1,314,616.00
3	Concrete block	472,044.45	0.155	73,166.89
4	Concrete precast	117,600.00	0.160	18,816.00
5	H-beam	97,000.00	0.98	95,351.00
6	Reinforced concrete piles	56,332.80	0.164	9238.58
7	Marble	35,324.32	0.210	7418.11
8	Aluminum, curtainwall	15,492.00	9.22	142,836.24
9	Glass	12,825.00	0.910	11,670.75
10	Steel pipe	10,711.50	1.45	15,531.68
11	Aluminum/trellis	9968.40	9.22	91,908.65
12	Moisture resistant fibrous plasterboard ceiling	6963.60	0.172	1197.74
13	Glass block	5175.00	0.910	4709.25
14	Angle cable trays	3745.32	1.46	5468.17
15	Heavy duty stainless steel angle	2732.22	5.51	15,054.53
16	Drainage cell with geofabric	2100.60	1.96	4117.18
17	PU elastomeric seamless membrane	1397.70	4.66	6513.29
18	Steel pipes	992.00	1.45	1438.40
19	ELV cable	800.00	2.79	2232.00
20	Water closet	560.00	1.61	901.60
21	Ductile iron pipe	342.00	2.03	694.26
22	Aluminum ventilation louver	220.00	9.22	2028.40
23	Access cover	150.00	2.03	304.50
24	Irrigation system – pump	100.00	3.10	310.00
25	Mirror	90.00	0.993	89.37
26	Paint	54.24	2.91	157.85
27	Hosereel electrical pump	33.80	2.23	75.37
28	Toilet roll holder	15.00	2.08	31.20
29	Robe hook	5.00	5.51	27.55
30	Air release valve	0.30	4.77	1.43
Total carbon footprint				4,064,699.89

The derivation of the emission factors in Table B was done manually by a practitioner through the process described by Fig. 1. The practitioner has to obtain the material composition of each item in the bill-of-material. After which, the practitioner has to match each single material of the item with a list of potential emission factors from the practitioner's database. This selection process is carried out to select the most appropriate and representative emission factor for each material. This process is performed based on the data quality requirement principles stated in the carbon footprint standards (ISO/TS 14067). The described procedure is a manual and laborious process. Once the selection of the emission factors are done, the final emission for each material is computed and aggregated to obtain the final embodied carbon footprint of the building materials.

To illustrate the process, we take item 6 (Reinforced Concrete Piles) as an example:

Total weight of item = 56,332.80 kg (refer to Table A)

Material composition of “Reinforced Concrete Piles”: 99% concrete, 1% steel (refer to Table A)

Number of emission factors of “concrete” in database: 79 (summarized in Table 2)

Number of emission factors of “steel” in database: 55 (summarized in Table 2)

At this step, there is an information gap regarding the specific type of concrete and steel that the item is composed of. Therefore, the practitioner is required to contact the activity data provider to obtain further specifications regarding the type of concrete (from concrete mix design) and steel (from specifications sheet or design drawings) used. This process is time-consuming as the practitioner has to obtain detailed information regarding all 30 items in the bill-of-materials.

After details of all the materials are obtained, manual matching of the appropriate emission factor to the materials can be made as follows.

Final emission factor selected for “concrete”: RC 40/50 (40/50 MPa), 30% Cement Replacement – Fly Ash, 0.155 kg CO₂e/kg

Final emission factor selected for “steel”: Rebar, 1.01 kg CO₂e/kg

Weighted average emission factor for “Reinforced Concrete Piles”

$$=0.155 \times 99\% + 1.01 \times 1\%$$

$$=0.164 \text{ kg CO}_2\text{e/kg}$$

$$\text{Emissions for “Reinforced Concrete Piles”} = 56,332.80 \times 0.164$$

$$=9238.58 \text{ kg CO}_2\text{e}$$

The above process is repeated for all the 30 items listed in the bill-of-materials.

Illustration of computation of carbon footprint using proposed technique

Table C
Estimated carbon footprint (pre-error correction).

No.	Item	Material class(es)	Total weight (kg)	Estimated carbon footprint (kg CO ₂ e)
1	Concrete, in-situ	Concrete	25,972,087.22	3,116,650.47
2	Rebar	Steel	1,301,600.00	2,381,928.00
3	Concrete block	Concrete	472,044.45	56,645.33
4	Concrete precast	Concrete	117,600.00	14,112.00
5	H-beam	Steel	97,000.00	177,510.00
6	Reinforced concrete piles	Steel, concrete	56,332.80	7723.23
7	Marble	Stone	35,324.32	3885.68
8	Aluminum, curtainwall	Aluminum	15,492.00	115,725.24
9	Glass	Glass	12,825.00	17,057.25
10	Steel pipe	Steel	10,711.50	19,602.05
11	Aluminum/trellis	Aluminum	9968.40	74,463.95
12	Moisture resistant fibrous plasterboard ceiling	Plaster, glass	6963.60	2371.80
13	Glass block	Glass	5175.00	6882.75
14	Angle cable trays	Steel	3745.32	6853.94
15	Heavy duty stainless steel angle	Stainless steel	2732.22	13,114.66
16	Drainage cell with geofabric	Plastic	2100.60	8297.37
17	PU elastomeric seamless membrane	Plastic	1397.70	5520.92
18	Steel pipes	Steel	992.00	1815.36
19	ELV cable	Copper, plastic	800.00	2462.40
20	Water closet	Ceramics	560.00	565.60
21	Ductile iron pipe	Steel	342.00	625.86
22	Aluminum ventilation louver	Aluminum	220.00	1643.40
23	Access cover	Steel	150.00	274.50
24	Irrigation system – pump	Plastics	100.00	395.00
25	Mirror	Glass, aluminum	90.00	125.23
26	Paint	Paint	54.24	110.11
27	Hosereel electrical pump	Steel, copper	33.80	72.30
28	Toilet roll holder	Plastics	15.00	59.25
29	Robe hook	Stainless steel	5.00	24.00
30	Air release valve	Other metals	0.30	1.10
Total estimated carbon footprint				6,036,518.73

The derivation of Table C was performed using the process described in Fig. 2. Similar to the current technique, the first step is to obtain the material composition of each item in the bill-of-material. After which, the practitioner will proceed to identify the closest material class, defined in Table 2, for each item. Then, the appropriate breakdown of the item weight is done for each material class, and the estimated emissions are calculated using the emission factor estimators in Table 2. After the emission of each item is estimated, the result is aggregated to obtain an initial estimate of the total carbon footprint.

To illustrate the process, we take item 6 (Reinforced Concrete Piles) as an example:

Total weight of item = 56,332.80 kg (refer to Table A)

Material composition of “Reinforced Concrete Piles”: 99% concrete, 1% steel (refer to Table A)

Material class for “concrete”: Concrete

Material class for “steel”: Steel

Emission factor estimator for material class “Concrete” = 0.119 kg CO₂e/kg

Emission factor estimator for material class “Steel” = 1.83 kg CO₂e/kg

Weighted emission factor estimator for “Reinforced Concrete Piles”

$$=0.12 \times 99\% + 1.83 \times 1\%$$

$$=0.1371 \text{ kg CO}_2\text{e/kg}$$

Estimated emissions for “Reinforced Concrete Piles” = 56,332.80 × 0.1371

$$=7723.23 \text{ kg CO}_2\text{e}$$

At this stage, it is apparent that the estimated total carbon footprint (6,036,518.73 kg CO₂e) deviates significantly from the actual carbon footprint (4,064,699.89 kg CO₂e) computed by the practitioner. The estimated carbon footprint is about 149% of the actual carbon footprint.

The error analysis and reduction procedure will identify the items with significant estimation errors to pin-point data items that require further data refinement. Table D shows a summary of the data items with the associated expected error index computed using equation (8).

To illustrate the process, we take item 6 (Reinforced Concrete Piles) to continue on as the example:

$$\text{Expected error index}_i = \frac{\sum_{j=1}^M w_{ij} \times s_{ij} \times P(\text{error})_{ij}}{\sum_{k=1}^N \sum_{l=1}^M w_{k,l} \times s_{k,l} \times P(\text{error})_{k,l}}$$

where:

w_{ij} is the weight of the j th material class for the i th item in the BOM

s_{ij} is the sample standard deviation of the j th material class for the i th item in the BOM (From Table 2)

$P(\text{error})$ is the expected probability of error of the j th material class for the i th item in the BOM (From Table 2)

$w_{k,l}$ is the weight of the k th material class for the l th item in the BOM

$s_{k,l}$ is the sample standard deviation of the k th material class for the l th item in the BOM

N is the total number of items in the BOM

M is the total number of material class

For item 6, there are only 2 material classes, therefore, the numerator can be computed as follows.

$$\sum_{j=1}^2 w_{ij} \times s_{ij} \times P(\text{error})_{ij}$$

$$= (56,332.80 \times 99\%) \times 0.045 \times 0.241 + (56,332.80 \times 1\%) \times 0.837 \times 0.417$$

$$= 801.44$$

Repeating the above to compute for all the 30 items, followed by aggregation will yield the denominator of the equation.

$$\sum_{k=1}^{30} \sum_{l=1}^M w_{k,l} \times s_{k,l} \times P(\text{error})_{k,l} = 888,031.89$$

Finally, the expected error index for reinforced concrete piles can be computed.

$$\text{Expected error index}_{\text{item 6}} = \frac{801.44}{88,8031.89} \times 100\% = 0.09\%$$

After all the expected error index are calculated for all the 30 items, each item is classified as high, medium or low error band based on the range defined in Table 1.

Table D
Expected error index for estimated carbon footprint.

No.	Item	Expected error index	Error band
1	Concrete, in-situ	31.9%	High
2	Rebar	51.2%	High
3	Concrete block	0.58%	Low
4	Concrete precast	0.14%	Low
5	H-beam	3.81%	Medium
6	Reinforced concrete piles	0.09%	Low
7	Marble	0.72%	Low
8	Aluminum, curtainwall	5.54%	Medium
9	Glass	0.66%	Low
10	Steel pipe	0.42%	Low
11	Aluminum/trellis	3.56%	Medium
12	Moisture resistant fibrous plasterboard ceiling	0.18%	Low
13	Glass block	0.26%	Low
14	Angle cable trays	0.15%	Low
15	Heavy duty stainless steel angle	0.11%	Low
16	Drainage cell with geofabric	0.27%	Low
17	PU elastomeric seamless membrane	0.18%	Low
18	Steel pipes	0.039%	Low
19	ELV cable	0.11%	Low
20	Water closet	0.035%	Low
21	Ductile iron pipe	0.013%	Low
22	Aluminum ventilation louver	0.079%	Low
23	Access cover	0.0059%	Low
24	Irrigation system – pump	0.013%	Low
25	Mirror	0.0049%	Low
26	Paint	0.0046%	Low
27	Hosereel electrical pump	0.0024%	Low
28	Toilet roll holder	0.002%	Low
29	Robe hook	0.0002%	Low
30	Air release valve	0.0001%	Low

Table E
Data items for further refinement.

No.	Item	Expected error index	Error band
1	Concrete, in-situ	31.9%	High
2	Rebar	51.2%	High
5	H-beam	3.81%	Medium
8	Aluminum, curtainwall	5.54%	Medium
11	Aluminum/trellis	3.56%	Medium

During the data refinement stage, the practitioner request for more information regarding the shortlisted items, marked as either “high” or “medium” in terms of error band. The practitioner can utilize the “high” or “medium” to prioritize the importance and urgency of the data to be refined. This step cuts down on the need to request for detailed data for all 30 items to only 5 items.

It is assumed that the carbon footprint for these 5 items will be the same as that derived from the conventional method as the practitioner undergoes identical process of data gathering and computation. Table F shows the carbon footprint after refinement by the practitioner.

Table F
Carbon footprint of the 5 items after data refinement process.

No.	Item	Material class(es)	Total weight (kg)	Carbon footprint after data refinement (kg CO ₂ e)
1	Concrete, in-situ	Concrete	25,972,087.22	2,238,793.92
2	Rebar	Steel	1,301,600.00	1,314,616.00
5	H-beam	Steel	97,000.00	95,351.00
8	Aluminum, curtainwall	Aluminum	15,492.00	142,836.24
11	Aluminum/trellis	Aluminum	9968.40	91,908.65

Finally, after updating the carbon footprint of the 5 items shown in Table F, the final estimated carbon footprint is compiled and shown in Table G.

Table G
Final estimated carbon footprint.

No.	Item	Material class(es)	Total weight (kg)	Revised estimated carbon footprint (kg CO ₂ e)
1	Concrete, in-situ	Concrete	25,972,087.22	2,238,793.92
2	Rebar	Steel	1,301,600.00	1,314,616.00
3	Concrete block	Concrete	472,044.45	56,645.33
4	Concrete precast	Concrete	117,600.00	14,112.00
5	H-beam	Steel	97,000.00	95,351.00
6	Reinforced concrete piles	Steel, concrete	56,332.80	7723.23
7	Marble	Stone	35,324.32	3885.68
8	Aluminum, curtainwall	Aluminum	15,492.00	142,836.24
9	Glass	Glass	12,825.00	17,057.25
10	Steel pipe	Steel	10,711.50	19,602.05
11	Aluminum/trellis	Aluminum	9968.40	91,908.65
12	Moisture resistant fibrous plasterboard ceiling	Plaster, glass	6963.60	2371.80
13	Glass block	Glass	5175.00	6882.75
14	Angle cable trays	Steel	3745.32	6853.94
15	Heavy duty stainless steel angle	Stainless steel	2732.22	13,114.66
16	Drainage cell with geofabric	Plastic	2100.60	8297.37
17	PU elastomeric seamless membrane	Plastic	1397.70	5520.92
18	Steel pipes	Steel	992.00	1815.36
19	ELV cable	Copper, plastic	800.00	2462.40
20	Water closet	Ceramics	560.00	565.60
21	Ductile iron pipe	Steel	342.00	625.86

(continued on next page)

Table G (continued)

No.	Item	Material class(es)	Total weight (kg)	Revised estimated carbon footprint (kg CO ₂ e)
22	Aluminum ventilation louver	Aluminum	220.00	1643.40
23	Access cover	Steel	150.00	274.50
24	Irrigation system – pump	Plastics	100.00	395.00
25	Mirror	Glass, aluminum	90.00	125.23
26	Paint	Paint	54.24	110.11
27	Hosereel electrical pump	Steel, copper	33.80	72.30
28	Toilet roll holder	Plastics	15.00	59.25
29	Robe hook	Stainless steel	5.00	24.00
30	Air release valve	Other metals	0.30	1.10
Total estimated carbon footprint				4,053,746.88

The bold entries represent those values that were updated after the error correction procedure.

Comparison of total carbon footprint computed using the two methods

Table H shows the comparison of the carbon footprint derived from the current technique performed manually by the practitioner compared to the proposed technique, assisted by a pre-computed list of emission factor estimators and material classes. The initial estimation produced by the proposed technique deviates significantly from the current technique (about 49% overestimated in this case). However, after going through the error estimation and correction process, the percentage error has markedly decreased to less than 1% in the final estimation.

Table H

Final estimated carbon footprint.

Computation method	Total carbon footprint (kg CO ₂ e)	Normalized total carbon footprint (current technique = 100)
Current technique	4,064,699.89	100.00
Proposed technique (initial estimation)	6,036,518.73	148.51
Proposed technique (final estimation)	4,053,746.88	99.73

References

Abanda, F.H., Tah, J.H.M., Cheung, F.K.T., 2013. Mathematical modelling of embodied energy, greenhouse gases, waste, time-cost parameters of building projects: a review. *Build. Environ.* 59, 23–37.

Akadiri, P.O., Olomolaiye, P.O., Chinyio, A.E., 2013. Multi-criteria evaluation model for the selection of sustainable materials for building projects. *Autom. Constr.* 30, 113–125.

Blue Angel, 1978. Blue Angel Website [Online]. Available at: <http://www.blauer-engel.de/> (accessed 08.07.14.).

BCA, 2012a. As-Built GM Submission before TOP for Non-Residential Buildings [Online]. Available at: http://www.bca.gov.sg/Envvsuslegislation/others/As-Built-Form-NonResBldg_3rd_Ed.pdf (accessed 07.07.14.).

BCA, 2012b. GM Submission at BP Stage for Residential Buildings [Online]. Available at: http://www.bca.gov.sg/Envvsuslegislation/others/Form-ResBldg_3rd_Ed.pdf (accessed 07.07.14.).

Bribián, I.Z., Capilla, A.V., Usón, A.A., 2011. Life cycle assessment of building materials: comparative analysis of energy and environmental impacts and evaluation of the eco-efficiency improvement potential. *Build. Environ.* 46 (5), 1133–1140.

BSI, 2011. PAS 2050:2011 Specification for the Assessment of the Life Cycle Greenhouse Gas Emissions of Goods and Services. British Standards Institution, London.

Buyle, M., Braet, J., Audenaert, A., 2013. Life cycle assessment in the construction sector: a review. *Renew. Sustain. Energy Rev.* 26, 379–388.

CECP, 1998. China Energy Conservation Program (CECP) Website [Online]. Available at: <http://www.cecp.org.cn/> (accessed 08.07.14.).

Centre for Life Cycle Inventories, 2010. Ecoinvent Lifecycle Database. Switzerland: Centre for Life Cycle Inventories, Zürich.

Cole, R.J., Rousseau, D., 1992. Environmental auditing for building construction: energy and air pollution indices for building materials. *Build. Environ.* 27 (1), 23–30.

Cooper, J.S., Fava, J.A., 2006. Life-cycle assessment practitioner survey. *J. Ind. Ecol.* 10 (4), 12–14.

DeCoursey, W., 2003. *Statistics and Probability for Engineering Applications*. Elsevier Science (USA), Massachusetts.

Delft University of Technology, 2008. IdeMat LCI Database. Delft, Netherlands.

Eaton, K.J., Amaton, A., 1998. A Comparative Environmental Life Cycle Assessment of Modern Office Buildings. s.l. Steel Construction Institute.

Energy Star, 1992. Energy Star – Green Buildings and Energy Star [Online]. Available at: <http://www.energystar.gov/buildings> (accessed 08.07.14.).

Engin, A., Frances, Y., 2009. Zero Carbon Isn't Really Zero: Why Embodied Carbon in Materials Can't Be Ignored [Online]. Available at: http://www.di.net/articles/zero_carbon/ (accessed 04.03.15.).

EPA, 1992. United States Environmental Protection Agency – Design for Environment Label [Online]. Available at: <http://www.epa.gov/dfe/pubs/projects/formulat/saferproductlabeling.htm> (accessed 07.07.14.).

ESU-services, 1996. ETH-ESU 96 Libraries. ESU-services, Zürich, Uster.

Franklin USA 98, 1998. Franklin USA LCI Library. Sylvatica, USA.

GECA, 2002. Good Environmental Choice Australia (GECA) [Online]. Available at: <http://www.geca.org.au/> (accessed 08.07.14.).

González, M.J., Navarro, J.G., 2006. Assessment of the decrease of CO₂ emissions in the construction field through the selection of materials: practical case study of three houses of low environmental impact. *Build. Environ.* 41 (7), 902–909.

Gursel, A.P., Maryman, H., Ostertag, C., Jun, 2015. “A life-cycle approach to environmental, mechanical, and durability properties of ‘green’ concrete mixes with rice husk ash.”. *J. Clean. Prod.* 112, 823–836.

Holtzhausen, H.J., 2007. Embodied energy and its impact on architectural decisions. *WIT Trans. Ecol. Environ.* 102, 377–385.

Huberman, N., Pearlmuter, D., 2008. A life-cycle energy analysis of building materials in the Negev desert. *Energy Build.* 40 (5), 837–848.

Ibn-Mohammed, T., et al., 2013. Operational vs. embodied emissions in buildings—a review of current trends. *Energy Build.* 66, 232–245.

IBU, 2004. IBU Type III Environmental Declaration (IBU Environmental Product Declaration) Website [Online]. Available at: <http://bau-umwelt.de/hp1/Institut-Bauen-und-Umwelt-e-V.htm> (accessed 08.07.14.).

International Energy Agency, 2013. Redrawing the Energy-Climate Map. International Energy Agency, Paris Cedex.

ISO, 2013. ISO/TS 14067:2013 Greenhouse Gases – Carbon Footprint of Products – Requirements and Guidelines for Quantification and Communication. International Organization for Standardization, Switzerland.

International Stainless Steel Forum, 2005. ISSF Life Cycle Inventory and Analysis. ISSF, Brussels.

Jönsson, Å., 1998. Life Cycle Assessment of Building Products. Case Studies and Methodology. Technical Environmental Planning, Chalmers University of Technology, Göteborg.

Khasreen, M.M., Banfill, P.F., Menzies, G.F., 2009. Life-cycle assessment and the environmental impact of buildings: a review. *Sustainability* 1 (3), 674–701.

Koskela, L., 1992. Application of the New Production Philosophy to Construction. Technical Report No. 72. Center for Integrated Facility Engineering, Department of Civil Engineering, Stanford University, Stanford.

Langston, Y.L., Langston, C.A., 2008. Reliability of building embodied energy modelling: an analysis of 30 Melbourne case studies. *Constr. Manag. Econ.* 26 (2), 147–160.

Lee, B., Trcka, M., Hensen, J.L., 2011. Embodied energy of building materials and green building rating systems—a case study for industrial halls. *Sustain. Cities Soc.* 1 (2), 67–71.

Lucon, O., et al., 2014. Buildings. In: *Climate Change 2014: Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, United Kingdom and New York, NY, USA.

Meinrenken, C.J., Kaufman, S.M., Ramesh, S., Lackner, K.S., 2012. Fast carbon footprinting for large product portfolios. *J. Ind. Ecol.* 16 (5), 669–679.

Montgomery, D.C., Runger, G.C., 2011. *Applied Statistics and Probability for Engineers*, fifth ed. John Wiley & Sons, Inc., California.

Nabel, B., Alcorn, A., Wittstock, B., 2008. Life Cycle Assessment: Adopting and Adapting Overseas LCA Data and Methodologies for Building Materials in New Zealand. Ministry of Agriculture and Forestry (New Zealand), Wellington.

Ng, R., et al., 2012. An algorithmic approach to streamlining product carbon footprint quantification: a case study on sheet metal parts. *Int. J. Autom. Technol.* 6 (3), 312–321.

Optis, M., Wild, P., 2010. Inadequate documentation in published life cycle energy reports on buildings. *Int. J. Life Cycle Assess.* 15 (7), 644–651.

PE International, 2010. PE International Lifecycle Database. PE International, Stuttgart.

Plank, R., 2008. The principles of sustainable construction. *IES J. Part A Civ. Struct. Eng.* 1 (4), 301–307.

Plastics Europe, 2005. *PlasticsEurope Eco-profile*. Plastics Europe, Brussels.

Ramesh, T., Prakash, R., Shukla, K.K., 2010. Life cycle energy analysis of buildings: an overview. *Energy Build.* 42 (10), 1592–1600.

- Reap, J., Roman, F., Duncan, S., Bras, B., 2008. A survey of unresolved problems in life cycle assessment. *Int. J. Life Cycle Assess.* 13 (5), 374–388.
- Rincón, L., 2011. Material Flow Analysis of the Building Sector in Lleida (PhD thesis). University of Lleida, Lleida, Spain.
- Rugrungruang, F., Song, B., Yeo, Z., Gwee, H.K., 2010. A Method for Applying LCA for Eco-Performance Improvement of Complex Products. Care Innovation, Vienna, Austria.
- Sahagun, D., Moncaster, A., 2012. How Much Do We Have to Spend to Save? Calculating the Embodied Carbon Costs of Retrofit [Online]. Available at: http://www.salford.ac.uk/_data/assets/pdf_file/0011/142400/036-Sahagun.pdf (accessed 08.07.14.).
- Sartori, I., Hestnes, A., 2007. Energy use in the life cycle of conventional and low-energy buildings: a review article. *Energy Build.* 39 (3), 249–257.
- SEC, 2014. Singapore Environment Council – Singapore Green Labelling Scheme [Online]. Available at: <http://www.sec.org.sg/sxls/> (accessed 07.07.14.).
- Sustainable Energy Research Team (SERT), 2011. Inventory of Carbon & Energy. University of Bath, Bath, UK.
- SGBC, 2014. Singapore Green Building Council – Green Certifications [Online]. Available at: <http://www.sgbc.sg/green-certifications/> (accessed 07.07.14.).
- Silvestre, J., de Brito, J., Pinheiro, M., 2014. Environmental impacts and benefits of the end-of-life of building materials – calculation rules, results and contribution to a “cradle to cradle” life cycle. *J. Clean. Prod.* 66, 37–45.
- SIMTech, 2010. Carbon Footprint of Green Construction Materials. SIMTech, Singapore.
- Singh, A., Berghorn, G., Joshi, S., Syal, M., 2011. Review of life-cycle assessment applications in building construction. *J. Archit. Eng.* 17 (1), 15–23.
- SIRIM, 1997. SIRIM Certified Website [Online]. Available at: <http://www.sirim-qas.com.my/> (accessed 08.07.14.).
- Suzuki, M., Oka, T., 1998. Estimation of life cycle energy consumption and CO₂ emission of office buildings in Japan. *Energy Build.* 28 (1), 33–41.
- Swiss Packaging Institute, 2000. BUWAL 250 Library. Swiss Packaging Institute, Switzerland.
- Teixeira, R., Pax, S., 2011. A survey of life cycle assessment practitioners with a focus on the agri-food sector. *J. Ind. Ecol.* 15 (6), 817–820.
- Thormark, C., 2002. A low energy building in a life cycle—its embodied energy, energy need for operation and recycling potential. *Build. Environ.* 37 (4), 429–435.
- Torgal, F., 2014. Eco-efficient construction and building materials research under the EU Framework Programme Horizon 2020. *Constr. Build. Mater.* 51, 151–162.
- UNEP, 2007. Buildings and Climate Change: Status, Challenges and Opportunities. United Nations Environment Programme, Paris.
- World Steel Association, 2000. World Steel Association Life Cycle Assessment. World Steel Association, Brussels.
- WRI, 2011. Product Life Cycle Accounting and Reporting Standard. GHG Protocol, Washington.
- Yeo, Z., et al., 2013. Method for Rapid Estimation of Carbon Footprint Involving Complex Building Inventory Data – a Case Study. Springer Singapore, Singapore.